EventTrace: Verifying Ordered Semantic Events for Reinforcement Fine-Tuning of Vision-Language Navigation

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Abstract

Reinforcement fine-tuning of vision-language navigation (VLN) policies commonly relies on destination success or similarity to a reference route. These signals leave an instruction's intermediate requirements unresolved: reaching the endpoint does not establish that the agent entered the requested room or passed the specified landmark. We study an online reward that represents an instruction as an ordered list of semantic events and verifies each event against observations before and after executed actions. A frozen vision-language model supplies a three-way verdict (complete, incomplete, uncertain); motion and stopping checks reject visually plausible but unexecuted events. Each event is rewarded at most once, and the process reward is bounded below the success bonus. We integrate EventTrace into multi-turn ActiveVLN/GRPO and compare it with destination-only training for 64 optimization steps across three matched seeds. All six checkpoints and the SFT initialization are evaluated on the same 1,839 R2R val-unseen episodes. EventTrace changes success rate by $+0.65$, $+0.44$, and -2.88 percentage points across seeds; its mean paired change is -0.60 points (sample standard deviation 1.98). A single-AI blind review of 49 selected transitions, replayed with training-time image encoding, finds that 7 of 15 verifier completion predictions match its labels. Two subsequent independent human reviewers agree on only 33 of these 49 cases; their 16 disagreements remain unadjudicated. The method is operational, but these results do not establish a navigation improvement or reliable semantic verification.

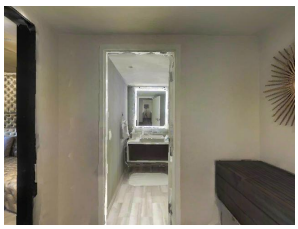
1. Introduction

Vision-language navigation asks an embodied agent to follow a natural-language route through visual observations. The Room-to-Room (R2R) task introduced instruction-guided navigation in real indoor environments [2]; VLN-CE moves this task into continuous control [8]. Success rate (SR) and success weighted by path length (SPL) capture destination attainment, yet a successful stop can coexist with skipped intermediate instructions. Instruction fidelity work made this gap explicit [7].

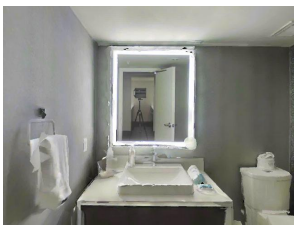
Recent reinforcement fine-tuning (RFT) approaches give visual-language policies online or trajectory-level feedback. ActiveVLN performs multi-turn exploration with group-relative policy optimization (GRPO) [16, 19]. VLN-R1 uses time-decayed action rewards [13]; Nav-R1 combines format, understanding, and navigation rewards [11]; SeeNav-Agent proposes step-level verifiable process rewards in another navigation benchmark [18]. Progress-Think learns semantic progress from visual histories and instruction prefixes [17]; SACA audits stepwise landmark progress and routes mixed-success and all-failure groups differently [9]. Our question is narrower: can a navigation instruction's ordered, physical events provide a bounded reward during closed-loop R2R-CE rollouts, without matching the policy's actions to a reference path?

We introduce EventTrace, a modular reward on top of an existing multi-turn policy. The reward uses a short event list, a frozen visual verifier with an abstention state, deterministic checks of motion or stopping, and a once-only order-preserving tracker. The actor sees only the original egocentric observations and instruction; simulator displacement is available only to the reward. Figure 1 illustrates the distinction between seeing a bathroom and actually entering it.

We evaluate the reward with matched three-seed training and complete R2R val-unseen coverage. The observed seed variation and an exploratory verifier audit limit the empirical claim: this implementation is reproducible but does not improve navigation consistently.



(a) Doorway view



(b) Interior view

Figure 1. Real simulator frames from a diagnostic replay of R2R training episode 2889. Visibility alone does not complete the event "enter bathroom." The verifier uses before/after views and executed motion; the ordered tracker later checks "stop near sink." These frames illustrate the mechanism and are not an independent accuracy test.

2. Related Work

Path and instruction fidelity. SR and SPL reward the destination and efficiency, whereas nDTW and coverage weighted by length score (CLS) compare trajectories with a reference route [6, 7]. Reference trajectories can be useful supervision but do not directly answer whether a distinct visual event actually occurred when multiple valid routes exist. Our reward targets event completion during execution; route metrics remain complementary evaluation measures.

RL for VLN. ActiveVLN is our implementation base because its environment can return feedback after each multi-action turn [19]. VLN-R1 and Nav-R1 show other ways to shape VLN rewards [11, 13]. Progress-Think aligns observation and instruction progress and co-trains a progress-aware policy [17]; SACA combines a landmark auditor with scenario-dependent group optimization [9]. SeeNav-Agent emphasizes step-level verification [18]. EventTrace tests a narrower combination of a once-only ordered event state, an explicit uncertainty outcome, and motion gates against rewarding a landmark merely seen without the requested transition. Our pilot applies GRPO to trajectory-level returns; it does not claim turn-level credit assignment or stronger navigation performance than these systems.

3. Method

3.1. Task and event representation

An instruction x yields a list $E(x) = (e_1, \ldots, e_K)$, where e_k contains a verb type, target phrase, and source span. Supported event types include enter, pass, descend, and stop_near. A frozen vision-language model (VLM) proposes at most three events, restricted to source spans in the instruction. The eight-episode wiring pilot uses curated lists; the main 64-step training runs parse instructions automatically at episode start. Parser errors produce an empty list and no semantic reward. We have not independently audited parser accuracy.

Table 1. Motion gates in the pilot. The VLM must also verify that the image pair supports the specified target.

Event	Count	Required gate
enter	3	Forward action; $\|\Delta p\| \geq 0.10$ m
pass	3	Forward action; $\|\Delta p\| \geq 0.10$ m
descend	3	Vertical change $\Delta y \leq -0.08$ m
stop_near	8	Executed STOP action

In the curated pilot, each of the eight episodes contains a stopping event; there are also three enter, three pass, and three descend events, for 17 in total. These counts describe the pilot distribution, not the prevalence of event types in R2R. Table 1 shows the deterministic portion of the verifier. Each gate can reject completion before a VLM call, but passing a gate is never sufficient to award a point.

At turn t, the policy receives the instruction, action history, and current egocentric image I_t, and emits up to three low-level actions. The simulator executes these actions and returns I_{t+1}, pose change Δp_t, and task metrics. Only the current pending event e_{k_t} is queried. The frozen verifier produces

$$v_t = V(e_{k_t}, I_t, I_{t+1}, a_t, \Delta p_t) \in \{\mathrm{Y}, \mathrm{N}, \mathrm{U}\}, \qquad (1)$$

where U denotes insufficient evidence. A view of a landmark is not by itself a completion verdict. For crossing and passing, a forward displacement check is required; descending requires a negative vertical displacement; stopping near a target requires an executed STOP. The VLM inspects whether the image pair supports the specified target and transition. These are deliberately conservative gates, not a learned proof of physical state.

3.2. Ordered, bounded process reward

Let $k_t \in \{1, \ldots, K+1\}$ index the first unfinished event. The event fires only when its verifier says Y. Then

$$c_t = \mathbb{1}[k_t \leq K \wedge v_t = \mathrm{Y}], \qquad (2)$$

$$k_{t+1} = k_t + c_t, \qquad r_t^{\mathrm{sem}} = \frac{\lambda}{K} c_t. \qquad (3)$$

A N or U verdict leaves k_t unchanged. The tracker prevents repeated reward for the same landmark, enforces event order, and guarantees $\sum_t r_t^{\mathrm{sem}} \leq \lambda$. Because only the pending event is checked, at most one event fires per turn even if the agent performs several

actions. This avoids ambiguous credit when a model emits an action batch but can delay recognition of adjacent events. The tracker retains a confirmed event after later image changes, so a completed event cannot be unconfirmed by viewpoint drift.

The base destination term is the weighted-success reward already used by ActiveVLN, with nDTW weight zero. We add a fixed success floor b to both arms:

$$R(\tau) = \sum_t r_t^{\text{sem}} + S(\tau)\left[b + \alpha\left(1 - \min\left(\frac{d_T}{3}, 1\right)\right)\right], \tag{4}$$

where $S(\tau)$ is simulator success after STOP, d_T is terminal distance in meters, $\alpha = 15$, $b = 2$, and $\lambda \in \{0, 1\}$. Hence any success receives at least 2 while a failure receives at most 1 semantic point. This priority holds for this nonnegative reward configuration; it does not imply that the verifier is correct. The bound also makes failure reward interpretable: partial event completion can distinguish two unsuccessful attempts without allowing a failed route to outrank a successful one solely through event bonuses. The STOP event is distinct from simulator success: it demands visual proximity to its instruction target, while $S(\tau)$ uses the simulator's goal region. Disagreement between these signals is precisely a case the proposed reward aims to expose.

For each training episode, we sample two trajectories and normalize their returns within the group for GRPO [16]. Event rewards are emitted after each turn, but the current trainer sums them before constructing a trajectory advantage. Consequently, the implementation changes which trajectories are preferred, not which individual turn receives the policy gradient. Turn-level event timing is available in the log for future credit-assignment work.

3.3. Verifier failure policy

The verifier is a frozen Qwen3-VL-8B-Instruct model. A service timeout, malformed response, or uncertain decision yields zero event reward; it does not end the episode. The reward requires no reference action prefix. Simulator pose is used only for reward verification, never added to the actor observation. In the pilot, the fixed event manifest is matched to both episode ID and exact instruction text to avoid assigning another episode's event list; the main runs use online parsing. The verifier sees two RGB frames resized to fit within 448×448 pixels, the event type and target, its source phrase, the executed action text, translation magnitude, and vertical displacement. Its output is a deterministic one-letter Y/N/U response; generated explanations are not treated as confidence scores. This makes the reward externally auditable through a saved per-turn verdict, although a forced-choice label is still fallible. The HTTP verifier is separated from the simulator so it can be frozen and logged independently of policy updates.

4. Experimental Setup

Environment and policy. We use R2R train in VLN-CE, rendered from Matterport3D scans [3], the ActiveVLN implementation, and an SFT-initialized Qwen2.5-VL-3B policy. Matterport3D data and derived imagery are subject to its academic terms of use. Each turn contains at most three grouped action commands. The configured limits are 12 turns and 36 commands; the strict budget check can permit one extra command, and a command may execute several simulator steps. Actor sampling uses temperature 1.2 and two independently sampled rollouts per item. GRPO trains on two A800 GPUs at learning rate 10^{-6} with batches of four episodes. The destination-only control sets $\lambda = 0$; the event arm sets $\lambda = 1$. Both arms have the same SFT initialization, destination reward, data order within seed, and optimizer budget.

Pilot and corrected main training. We first replay eight training episodes with expert actions solely to inspect expected events, then run a two-step reward-wiring pilot with curated events. These replays are neither held-out labels nor training results for the main comparison. For the main study, each arm runs 64 optimization steps at seeds 11, 22, and 33: 256 train-episode exposures and 512 sampled rollouts per arm and seed. The event arm uses automatically parsed events. A preflight check detected that a fixed per-request sampling seed duplicated every two-rollout GRPO group and yielded zero actor gradients; we excluded that attempt, removed the per-request seed while retaining the engine seed, and reran all six models. Every corrected group has distinct sampled actions. The archive records the excluded attempt and corrected patch.

Full val-unseen evaluation. SR is the fraction of trajectories satisfying the simulator goal condition; SPL also accounts for path efficiency. Mean terminal goal distance is in meters. Training rollout success and return are diagnostics, not held-out outcomes; returns differ in definition between arms. The SFT model and six corrected step-64 checkpoints run on the identical manifest of all 1,839 R2R val-unseen episodes from 11 scenes. We use one generation per episode, temperature 0.2, top-p 0.8, a 12-turn limit, and at most 76,800 image pixels. Four Habitat shards per checkpoint divide the manifest without overlap. The evaluator uses

Table 2. Observed pilot results on eight training episodes. Each arm ran two optimizer steps and produced 16 sampled rollouts. The event arm's larger mean return is expected from the added reward and does not indicate better navigation.

Metric	Destination only	EventTrace
Successful rollouts	13/16	13/16
Mean trajectory return	9.325	10.455
Mean event progress	–	0.656
Confirmed event hits	–	23
Uncertain verifier turns	–	33
Verifier service errors	–	0

the same prompt and environment-specific OpenAI-compatible request and image-resize shim for all seven models. The completion gate checks the exact episode-ID set and zero inference errors. Episode IDs are paired within each seed, but stochastic decoding uses only one generation per checkpoint and is an additional source of variation.

Blind verifier review. Separately, we collect shortest-path follower trajectories from 16 different val-unseen episodes, of which 12 yield at least one automatically parsed event. Before querying the verifier, one AI-assisted reviewer inspects RGB image pairs, executed actions, and motion and fixes 49 three-way completion labels across those 12 episodes. For the reported replay, both frames use the training client's 448-pixel JPEG encoding. The sample includes apparent transitions, controls, and final stops. The original AI review was independent of verifier predictions but had no human ground truth. Subsequently, two independent humans labeled the same answer-free, randomized 49-item package without access to predictions or prior labels. Their files were locked before comparison. We report their agreement separately and do not treat unresolved labels as verifier ground truth. Selected class proportions are not representative of deployment traffic.

5. Results and Analysis

Reward reaches optimization. In the first optimizer step, both arms sample the same eight action sequences and attain seven successes. For every matched trajectory, the difference in total return equals its semantic event score, including a failed episode 7110 trajectory that receives 0.5 for completing "descend stairs" but never reaches the destination. The modified return is consumed by the training reward manager, and both arms save step-2 checkpoints after nonzero policy updates. This verifies the wiring of Eq. 4; it is not evidence that event reward improves the policy.

What the verifier recognizes. In diagnostic expert replays, four of eight episodes receive full event credit, two receive partial credit, and two receive none; the tracker confirms 11 of 17 curated events. For episode 2889, seeing the sink from outside the bathroom produces uncertain verdicts before entrance, while later motion and view changes confirm enter. A rotation followed by premature STOP yields zero semantic credit in a negative replay. There are also false negatives: one shallow stair descent is rejected by the vertical-motion threshold, and one landmark is never confirmed. Because the events were curated using these episodes and the replays lack independent annotations, 11/17 must not be reported as verifier recall or accuracy.

Main training diagnostics. All six corrected runs finish 64 steps, with 256 diverse GRPO groups per arm and seed. Training-rollout success is 111/512 versus 119/512 (control versus event) for seed 11, 108/512 versus 119/512 for seed 22, and 88/512 versus 82/512 for seed 33. These on-policy samples change as training proceeds, so they are not comparable to the held-out evaluation. The event verifier records 272, 275, and 243 confirmed event hits over the three event-arm runs. These counts are not event accuracy: events are automatically parsed and no independent labels cover these rollouts.

Complete val-unseen results. Table 3 gives all seven checkpoints on the same 1,839 episodes. Within matched seeds, EventTrace changes SR by $+0.65$, $+0.44$, and -2.88 percentage points, and SPL by $+0.58$, $+0.49$, and -3.49 points. The mean paired seed change is -0.60 ± 1.98 points for SR and -0.80 ± 2.32 points for SPL (mean $\pm$ sample standard deviation over three seeds). At the episode level, event-only versus control-only successes are 157 versus 145, 197 versus 189, and 155 versus 208. This variation does not support a consistent improvement. All seven evaluations have exact 1,839-episode coverage and zero logged inference errors. The earlier 16-episode, two-step probe yielded 6/16 successes for every arm; it was exploratory and is superseded by the complete comparison.

Verifier failure analysis. Table 4 compares the frozen verifier with labels fixed before its predictions were obtained. Seven of eight reviewed completions receive Y, but only 7 of 15 Y predictions match a reviewed completion; four are on clear non-completions and four on ambiguous turns. One false positive occurs after the

Table 3. Complete R2R val-unseen evaluation. Every row uses the same 1,839 episodes from 11 scenes, one generation per episode, and zero inference errors. Goal distance is in meters. C: destination-only control; E: EventTrace.

Checkpoint	Successes	SR (%)	SPL (%)	Dist.
SFT	544	29.58	28.69	7.03
Seed 11 C	479	26.05	25.32	7.75
Seed 11 E	491	26.70	25.91	7.63
Seed 22 C	566	30.78	30.14	6.91
Seed 22 E	574	31.21	30.63	6.98
Seed 33 C	502	27.30	26.65	7.14
Seed 33 E	449	24.42	23.16	7.76

Table 4. Training-encoded replay of 49 selected val-unseen event transitions from 12 episodes. Rows are blind review labels and columns are frozen verifier predictions. U means evidence insufficient; the single-AI review has no independent human adjudication.

	Pred. Y	Pred. N	Pred. U
Review Y	7	0	1
Review N	4	13	12
Review U	4	1	7

agent has already entered the requested room, exposing a failure to distinguish current state from an event newly completed this turn. Motion gates alone do not eliminate this error. These selected examples cannot establish deployment precision, but they make verifier reliability a concrete limitation of the present reward.

Independent human review. The two blinded human reviewers agree on 33/49 three-way labels (67.35%; Cohen's $\kappa = 0.459$). Agreement is 17/23 for enter, 6/10 for pass, and 10/16 for stop_near; 16 cases remain disputed. The brief rationales include interpretations of motion toward a landmark as completion and cases where a previously visited landmark cannot be established from the supplied before/after views. This shows that the local evidence and event rubric can be underdetermined; it does not isolate model error from label ambiguity. No model-versus-human precision or recall is computed before independent adjudication, and this deliberately selected pilot cannot estimate population accuracy.

6. Discussion and Limitations

The main risk is reward misspecification: the VLM may confuse doorways, and displacement gates may miss small valid movements. One-at-a-time verification may miss events completed together. Before/after frames limit timing to a turn; trajectory-level GRPO offers limited credit assignment to the decisive action.

The event tracker assumes a single valid execution order. We have not independently checked whether the parsed sequence matches the instruction's temporal constraints. The local verifier can miss completion that requires history or occurs in an alternative order consistent with the instruction. The frequency and navigation effect of these cases remain unmeasured.

The complete val-unseen comparison shows a negative mean effect and large seed variation, but 64 steps expose only 256 training-episode instances per run. For EventTrace, a larger training budget and repeated decoding have not been evaluated. The selected 49-example review now has two independent human labelers, but their disagreements have not been adjudicated; instruction parsing is also unaudited. We cannot isolate noisy verification, parser errors, reward scale, budget, or optimization as the cause. Separately sampled and adjudicated verifier and parser sets, followed by ablations of ordering, abstention, gates, reward budget, and turn-level advantage, are needed. This audit cannot establish event precision or recall.

6.1. Evidence from subsequent training changes

The follow-ups change credit assignment, training-state distribution, rollout group size, or the reward representation. They therefore have separate matched controls and development protocols; they are not ablations that identify why EventTrace failed. Table 5 summarizes the completed three-seed comparisons. Increasing rollout count gives a positive mean with additional simulator work, whereas policy-prefix training has a negative mean. Privileged turn-wise credit and normalized terminal-RLOO have positive means but opposite seed effects and exploratory intervals spanning zero. None validates an observation-only semantic reward. Detailed protocols, unsuccessful screens, and individual seed results are retained in the supplementary material.

The one-seed PPO/GAE implementation also regressed on val-seen. Its recurrence discounts generated action-text tokens, so this result does not evaluate decision-clock GAE or establish that actor–critic methods are unsuitable for VLN. Observation-only reward probes improved some offline rankings but failed online navigation or conservative calibration gates. These outcomes motivate separating evidence quality from the policy update rule rather than increasing group size alone.

Greedy-baseline ReMax pilot. A separate group-four ReMax-style update [10] run used one greedy baseline trajectory and four stochastic samples per prompt for

Table 5. Completed full val-unseen comparisons, each on 1,839 episodes. Entries are paired mean ± sample SD over three training seeds, in percentage points. Each row has its own matched control. EventTrace uses 64 steps and two rollouts per group; the follow-ups use 128 steps, with two rollouts for policy-prefix training and four for the last three rows. The group-size comparison uses a two-rollout control and twice its rollout compute. Differences between rows cannot rank methods.

Comparison	Δ SR	Δ SPL
EventTrace	-0.60 ± 1.98	-0.80 ± 2.32
Policy-prefix	-1.14 ± 5.41	-0.55 ± 5.75
Group four versus two	$+2.10 \pm 1.93$	$+2.51 \pm 2.27$
Privileged turn-wise	$+2.47 \pm 3.18$	$+2.51 \pm 2.78$
Terminal-RLOO	$+1.61 \pm 4.22$	$+1.07 \pm 3.60$

64 total optimizer updates. Against the previously evaluated, same-precision SFT reference on the fixed 256-episode, eight-scene screen, it achieved 39 successes versus 91, with paired changes of −20.31 SR and −19.22 SPL points. Both exports have exact coverage and zero inference errors, and an independent compact recount agrees. The fixed joint +2-point gate failed, so this single-seed candidate was not scaled. The screen reuses train scenes and is not a clean generalization test.

6.2. Positive trajectory weighting and completed replication

Positive-only trajectory update. After the inconsistent terminal-RLOO replication, we froze a different on-policy update on 45 R2R-train fit scenes, with eight development and eight reserved scenes. Both arms receive the same distance-weighted success-15 plus reference-route nDTW-5 terminal score R_i. For four sampled trajectories, the candidate uses

$$Q_i = R_i + 15S_i,$$
$$A_i = M_i \min\left\{1.5, \frac{[Q_i - \frac{1}{3}\sum_{j\neq i} Q_j]_+}{5}\right\}, \quad (5)$$

where S_i is simulator success and M_i is one only for successful trajectories or failures with $R_i \geq 2.5$. This prioritizes success and removes negative trajectory credit, inspired by self-imitation learning [12]. The update uses fresh on-policy groups and a peer-return baseline, without a replay buffer or learned value function. Each weight is broadcast to action tokens, with observation tokens masked. The matched GRPO control and candidate use the same SFT initialization, 512 fit rows, 64 steps, group size four, one policy epoch, sequence-mean loss aggregation, and reference KL coefficient 0.01. Every step has a nonzero actor gradient in both arms.

Model	Successes	SR (%)	SPL (%)
Unchanged SFT, FP16	555	30.18	29.23
GRPO, seed 11	223	12.13	12.12
Positive, seed 11	243	13.21	13.19
GRPO, seed 22	216	11.75	11.74
Positive, seed 22	220	11.96	11.92
GRPO, seed 33	217	11.80	11.78
Positive, seed 33	254	13.81	13.75

Table 6. Completed positive-trajectory follow-up on the same 1,839 val-unseen episodes. All engines use FP16 and the same decode settings. One unchanged SFT reference is shared across configured-seed replications.

This comparison tests success prioritization, eligibility filtering, fixed scaling and capping, and removal of negative trajectory credit jointly. Contributions of individual components remain unmeasured.

On the frozen 256-episode, eight-scene development screen, candidate/control successes are 51/17 (SR 19.92%/6.64%); paired SR and SPL both improve by 13.28 percentage points. Both arms have exact coverage and zero inference errors, and local independent recount agrees. A CPU navmesh reconstruction matches every raw SPL: all successful paths are shorter than the initial goal distance, so their SPL is one under the 3 m success region. Descriptive scene-bootstrap intervals are [8.71, 19.35] points for both differences. The first evaluation's wrong nDTW reference split was corrected before this complete sweep; all partial outputs and failed logs were archived. The original joint +2-point gate passed, and all six fresh matched 512-row, 128-step, group-four runs at configured seeds 11, 22, and 33 have now completed. Each arm records 1,024 episode exposures and 4,096 trajectories across two complete passes over the fixed 512 fit IDs. All 128 stored actor gradients and KL metrics are finite and nonzero. Training diagnostics are separate from the navigation evaluation below.

The reserved 256-episode screen has candidate/control successes 35/24, 23/23, and 30/27. Mean paired SR/SPL changes are +1.82/ + 1.76 points; descriptive configured-seed/scene intervals cross zero. Every final checkpoint was then evaluated on all 1,839 val-unseen episodes regardless of its reserved score. Coverage is exact and final inference errors are zero. Table 6 shows a small mean advantage over the updated GRPO control, +1.11 SR and +1.07 SPL points, rather than the large development difference.

Comparison to initialization. A separately frozen reference evaluation uses the unchanged SFT checkpoint in explicit FP16, matching the actual trained-

model engine precision. Its development successes are 109/256, versus the pilot candidate's 51/256: the candidate is 22.66 SR points below initialization despite beating its weak updated control. The shared full-split SFT obtains SR 30.18% and SPL 29.23%; the candidate mean is 13.00%/12.95%, or −17.18/−16.28 points versus SFT. Independent local recounts reconcile all episode exports, precision proofs, paired metrics, and descriptive intervals. Native BF16 SFT diagnostics remain separately archived. Thus this update does not improve on initialization and is not expanded further. The comparison does not identify the cause of degradation or the effects of individual components.

The agent clears request seeds; the frozen manager seeds GPU generation at 1000 + DP rank. Checked seed-11/22 starts share all 32 first-batch action/reward signatures. Independence of all rollout sampling streams is unestablished; configured-seed summaries are exploratory descriptions. The SFT reference is one decode, not three independent SFT training seeds. The extra evaluation was frozen after the pilot and before scale outputs. Reserved scenes may have been exposed in earlier SFT or exploratory work, and val-unseen has been used adaptively. This follow-up neither establishes clean unseen-scene benefit nor validates a deployable semantic verifier; its reward uses privileged simulator success and a reference route.

7. Conclusion

We formulated a path-free semantic process reward for closed-loop VLN that verifies ordered instruction events against executed motion and before/after views. A wiring pilot confirms that the bounded reward enters online GRPO. In three matched 64-step training runs and complete 1,839-episode val-unseen evaluation, EventTrace has a mean paired SR change of −0.60 percentage points relative to destination-only training, with inconsistent seed effects. Two independent human reviewers agree on 33/49 selected transitions, but the remaining cases lack adjudication and the selected sample cannot establish verifier accuracy. The current method is reproducible; reliable semantic verification and a navigation benefit remain unproven.

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EventTrace: Verifying Ordered Semantic Events for Reinforcement Fine-Tuning of Vision-Language Navigation

Supplementary Material

A. Scope of the supplementary evidence

This material preserves the exploratory protocols, failed screens, training audits, and post-result diagnostics summarized in the main text. All reported navigation differences have episode-matched controls and explicit coverage checks. The screens differ in episode sets, budgets, and prior use; their absolute success counts cannot rank methods across experiments. No unadjudicated semantic labels or offline reward rankings are treated as navigation gains.

Budget conventions in the positive-trajectory follow-up. A post-hoc source and log audit finds that training budget exhaustion unconditionally assigns task failure and zero outcome reward, including nDTW; evaluation instead executes STOP at the turn limit before collecting metrics. In the completed seed-11 control and candidate, respectively, 230 and 314 of 4,096 fit rollouts end strictly within 3 m but receive zero timeout reward. Both runs' 8,192 records and their original budget identities pass independent compact recount. In the prior full FP16 SFT evaluation, 238 of its 555 successes follow forced turn-limit STOP, among 1,083 such terminations. These are distinct training and evaluation trajectories; their counts are not counterfactual repaired SR. The checked base source is also shared by the pending three-direction pilot. This objective-convention difference motivates separately matched time-limit-aware credit tests, without changing existing frozen runs, metrics or gates. It does not identify a cause of degradation.

B. Credit-estimator and group-size development

Turn-level credit assignment pilot. We tested a group-four, actor-only return-to-go/leave-one-out estimator inspired by RLOO [1] and REINFORCE++ [4]. Each executed turn receives its remaining return relative to active peer rollouts; observation tokens are masked. The candidate also adds training-only simulator geodesic progress, so both credit assignment and reward change. From the same SFT state and 256 train rows, it and a destination-only GRPO control ran 64 steps; all candidate actor gradients were nonzero. On a frozen 256-episode val-seen manifest (53 scenes, exact coverage, zero inference errors), candidate/control successes were 68/78: paired SR -3.91 and SPL -3.72 points. The prespecified joint +2-point gate failed, so this mechanism was not expanded. One seed and decode cannot isolate turn-level credit assignment or establish unseen-scene gains. This is not a deployable semantic reward.

Credit–reward factorial follow-up. To separate the two changes in that pilot, we completed a four-arm, group-four comparison from the same SFT initialization and seed, with 256 training rows and 64 optimizer steps per arm. The arms cross trajectory-level GRPO versus turn-level leave-one-out updates with terminal-only versus additional training-time geodesic progress. On a new, disjoint 256-episode val-seen screen (50 scenes; complete coverage and zero inference errors), the outcome-only GRPO control succeeds on 97 episodes. The combined turn/progress, terminal-only turn, and progress-augmented GRPO arms succeed on 96, 98, and 106, respectively. Their paired SR/SPL changes against control are $(-0.39, -0.14)$, $(+0.39, +0.68)$, and $(+3.52, +1.51)$ percentage points. Although progress-augmented GRPO shows a positive SR difference, its SPL difference misses the frozen joint +2-point advancement gate. Descriptive scene-cluster 95% intervals for that arm span zero: $[-2.24, 9.26]$ SR and $[-4.21, 7.17]$ SPL points. An independent per-episode recount agrees with the paired analysis after a documented null-ID schema repair to the verifier; the reward, evaluator, and metric formulas were unchanged. These one-seed development results do not establish unseen-scene benefit, and the turn-level and trajectory-level rules also differ in effective token update scale. The process signal uses privileged geodesic distance, not the semantic event verifier.

Token-scale-matched terminal credit. The factorial's turn-level estimator divided each turn contrast by its action-token count, whereas the trajectory GRPO control applied an order-one advantage to each action token. We therefore tested a new group-four terminal-outcome-only leave-one-out estimator that standardizes the active-peer contrast and broadcasts it to each action token in the turn. From the same SFT initialization, it ran 64 steps on 256 train rows. On a third, disjoint 256-episode val-seen development screen (38 scenes), candidate/control successes are 108/103;

paired SR and SPL are +1.95 and +1.26 percentage points. Each arm covers every episode once with no inference errors, and an independent raw recount agrees. Exploratory scene-cluster intervals span zero: $[-4.50, 8.04]$ SR and $[-5.11, 7.26]$ SPL points. Both improvements fall short of the predeclared joint +2-point gate, so the original conditional three-seed expansion was skipped. A subsequent user-directed, explicitly post-result group-eight sensitivity and group-four three-seed expansion were conducted separately. This one-seed screen does not establish unseen-scene benefit or validate a semantic reward.

Post-result group-size sensitivity. Keeping the 64-step, 256-row prompts and terminal-RLOO rule fixed, we sampled eight rather than four trajectories per instruction. This doubles simulator rollout work. The group-eight run had logged nonzero actor gradients on 63 of 64 steps; the remaining step had an all-failure group and zero advantage. On the same 256-episode val-seen screen, the group-eight checkpoint succeeds on 112 episodes, versus 103 for the destination-only control and 108 for the group-four checkpoint. Its paired SR/SPL differences are +3.52/+2.82 points versus control and +1.56/ + 1.56 points versus group four. Every arm covers all 256 episodes across 38 scenes, with no inference errors, and an independent raw-stat recount agrees with the compact episode records. Descriptive scene-cluster intervals for group eight versus control span zero: $[-4.92, 12.13]$ SR and $[-5.28, 11.10]$ SPL points. This comparison was requested after the group-four advancement gate failed; the group-eight result does not retroactively satisfy that gate. It is one-seed, adaptively selected development evidence, with group size confounded by rollout compute.

Post-result group-four scale. The separately authorized expansion used the terminal-RLOO rule with group size four, 512 training rows, and 128 optimizer steps for each of seeds 11, 22, and 33. Every candidate and its same-seed destination-only control covered all 1,839 val-unseen episodes (11 scenes), with zero inference errors. Paired candidate-minus-control SR/SPL differences are +3.97/ + 3.30, −3.26/ − 3.07, and +4.13/ + 3.00 percentage points, respectively. Their mean differences are +1.61/ + 1.07 points, with seed standard deviations 4.22/3.60 points. Exploratory seed-and-scene intervals span zero: $[-3.03, 5.12]$ for SR and $[-2.86, 4.21]$ for SPL. The compact per-episode recount agrees with the raw-stat verifier. The opposite sign for seed 22 and the intervals preclude a robust-benefit claim. Val-unseen had already influenced method development, so this is larger development replication rather than an untouched test. It does not establish that the group-eight pilot will generalize or scale.

Outcome-consistent truncation follow-up. An $n = 4$, 64-step RLOO pilot jointly corrected budget truncation and restricted nDTW to successful episodes. It scored 97/256 versus SFT 109 and prior RLOO 105; paired SR/SPL changes were −4.69/ − 4.41 and −3.13/−3.88 points. Both frozen +2 gates failed; exact coverage, zero inference errors, and the raw-ID audit were verified. This one-seed, reused development result is not a generalization test; the bundled changes are not separately attributable.

Executed-turn-mask RLOO follow-up. A matched group-four, 64-update run omitted the final generated response from return and policy credit when turn exhaustion forced STOP without executing an action. It achieved 103/256 successes (SR 40.23%, SPL 39.48%), below the matched FP16 SFT reference (109; 42.58%, 41.26%) and prior terminal RLOO (105; 41.02%, 40.73%). Paired SR/SPL changes were −2.34/ − 1.78 and −0.78/ − 1.25 points, respectively. All 256 episodes across eight scenes were covered with zero inference errors; raw IDs and independent paired recounts agree. Both frozen +2/ + 2 gates failed, so it was not scaled. The evaluator wrapper's post-inference marker and alias errors were recovered with CPU-only processing; failure evidence is preserved. This single configured-seed result uses adaptively reused train-scene development episodes and is not a clean generalization test.

Multimodal GAE actor–critic pilot. We also tested a PPO actor [15] with masked token-level GAE [14], a separately initialized multimodal value head, and terminal-outcome-only feedback. The group-four actor and critic started from the same SFT model; 64 optimizer steps on 256 train rows had finite value losses and nonzero actor and critic gradients at every step. On an adaptively reused 778-episode val-seen development screen (53 scenes), the GAE candidate succeeds on 22 episodes versus 272 for the same-seed destination-only control: paired SR −32.13 and SPL −31.56 percentage points. Both arms have exact episode coverage and zero inference errors; two independent raw-stat recounts and a compact export recount agree. Descriptive scene-cluster intervals are $[-36.83, -27.43]$ SR and $[-36.19, -26.95]$ SPL points. Mean path lengths are 2.79 versus 6.53 m, consistent with shorter candidate trajectories but insufficient to

identify why learning failed. The prespecified joint +2-point scale gate failed, so the conditional three-seed GAE expansion was skipped. This one-seed development result is not evidence about unseen-scene generalization. The frozen recurrence discounts each generated action-text token ($\gamma = 0.99$, $\lambda = 0.95$), skipping observation spans. A post-result synthetic zero-value check shows text-length-dependent early credit at a fixed decision count. Decision-clock GAE remains untested, and this diagnostic does not identify the cause of the navigation decline.

Greedy-baseline ReMax-style pilot. The isolated run compares one greedy, nonsampled baseline response with four sampled responses per prompt, using the same Qwen2.5-VL-3B initialization, 512 fit rows, group size four, configured seed 11, and KL coefficient 0.1. Sixty-four optimizer updates comprise a two-update smoke run plus 62 resumed updates; the combined audit cross-checks all 64 TensorBoard metric steps against the resumed console log where available. The fixed reserved manifest contains 256 episodes across eight train scenes and was already used for SFT and development. ReMax succeeds on 39 episodes, versus 91 for the same-precision SFT reference; paired SR/SPL changes are $-20.31/-19.22$ points. The manifest coverage, zero inference errors, actual FP16 engine, checkpoint payload metadata, and independent compact recount pass. The joint +2-point gate fails; no scale-up follows. The raw audit trail and compact episode records are archived at experiments/three_directions/remax64/. Two pre-inference orchestration attempts are retained; neither opened candidate inference. This single configured-seed result on a reused screen neither ranks it against other development sets nor establishes a generalization effect.

C. Additional group-four development pilots

HAPO-style temporal-credit adaptation. Motivated by the horizon-adaptive temporal credit estimator in LongNav-R1 [5], we tested a group-specific Gaussian kernel over turn index ($\sigma = 2$) on the existing privileged geodesic-progress reward. This is a narrow adaptation, not a reproduction of the cited system. The configured-seed-11 run completed 64 updates on 512 fit rows; all updates had nonzero actor gradients. On the fixed, adaptively reused development256 manifest, the candidate achieved 104/256 successes versus 109 for the matched FP16 SFT reference. Paired changes were -1.95 SR and -1.56 SPL points. Coverage was exact across eight scenes, inference errors were zero, and an independent compact recount agreed. The scene-bootstrap intervals include zero. The frozen $+2/+2$ gate failed; no scale-up followed. Its reward uses privileged simulator geometry and is not a deployable semantic signal. The exact deployed source snapshot and all reports are archived under experiments/three_directions/hapo_n4_20261010/.

SFT-anchored RLOO with stronger KL. We also trained a group-four RLOO candidate from the same navigation SFT checkpoint with KL coefficient 1.0, holding the other frozen training settings and the 64-update budget fixed. The resumed audit covers the two-step smoke plus 62 resumed updates; all 64 actor-gradient steps were nonzero and KL values finite. On the same development256 manifest, it achieved 101/256 successes versus 109 for the matched FP16 SFT reference, with paired changes of -3.13 SR and -2.98 SPL points. Exact coverage, zero inference errors, and an independent compact recount passed. The wrapper initially failed to write a completion sentinel after inference; we preserved the failed exit and separately audited raw coverage before resuming CPU-only analysis, without rerunning model inference. The final verified report passes the raw/compact checks, but the frozen $+2/+2$ gate failed and the run was not scaled. The result does not establish that a larger KL coefficient caused lower performance.

Topological macro-action GRPO. As a structurally different policy, we trained ETP-R1's topological waypoint policy with group-four GRPO from its R2R DAgger initialization for 64 updates on 512 fit episodes. The separately audited two-update smoke is not counted as part of the 64 updates. On the adaptively reused 256-episode, eight-scene development screen, the candidate succeeds on 53 episodes (SR 20.70%, SPL 16.02%), compared with 35 (13.67%, 7.80%) for the same ETP DAgger initialization; paired changes are +7.03 SR and +8.22 SPL points. On those same episodes, the existing FP16 Qwen2.5-VL SFT reference succeeds on 109 (42.58%, 41.26%), leaving paired candidate-minus-SFT changes of -21.88 SR and -25.25 SPL points. Thus the candidate improves on its ETP initialization but fails the frozen joint $+2/+2$-point gate against the SFT reference, so no multi-seed expansion was started. This single configured-seed development comparison is adaptively reused and does not establish unseen-scene benefit or isolate algorithmic effects across architectures.

Rank-16 LoRA-RLOO. A separate group-four RLOO pilot updated rank-16 adapters on q- and v-projections of the same Qwen2.5-VL-3B navigation SFT initialization for 64 global optimizer steps, including its au-

dited two-step smoke. On the frozen 256-episode, eight-scene development screen, the FP16 candidate succeeds on 95 episodes, compared with 109 for the matched FP16 SFT reference and 47 for the matched destination-only GRPO anchor. Paired SR/SPL differences are $-5.47/-5.34$ percentage points versus SFT and $+18.75/+17.56$ points versus GRPO. The improvement over the weaker GRPO anchor does not offset regression against SFT: the frozen rule requires at least +2 points on both metrics against both references, so the method was not expanded. All 256 manifest episodes were evaluated with zero inference errors, and an independent compact recount agrees. This is one configured-seed result on adaptively reused R2R-train scenes, not a clean generalization test. The compact records, source identities, and verifier are archived with the supplementary experiment records.

Expert-aligned prefix imitation with outcome GRPO. We tested a token-level auxiliary loss on the generated action prefix that exactly matched the reference trajectory, stopping the mask at the first mismatch. The pilot used 64 group-four updates from the same SFT initialization. The auxiliary mask was active in 17 of 62 resumed updates. On the same reused 256-episode, eight-scene development screen, it achieved 41 successes versus 109 for matched FP16 SFT, with paired changes of −26.56 SR and −25.25 SPL points; both scene-bootstrap intervals were below zero. Episode coverage was exact, inference errors were zero, and independent raw and local compact recounts agreed. The frozen +2/+2 gate failed, so no scale-up followed. This is a single configured-seed result on adaptively reused training scenes and does not establish that sparse prefix imitation caused the decline or generalizes.

Turn-level RLOO and SRGPO-style process grouping. We compared three group-four methods initialized from the same navigation SFT: a terminal-outcome GRPO anchor, turn-level return-to-go/leave-one-out, and an SRGPO-style composite with shuffled cross-state process groups. Each used 64 updates on the same 512 fit episodes with configured seed 11. On the fixed, adaptively reused development256 screen, they achieved 47, 105, and 77 successes, respectively, versus 109 for the reused FP16 SFT reference. Paired SR/SPL changes versus SFT were $-24.22/-22.90$, $-1.56/-0.53$, and $-12.50/-11.32$ points. RLOO and the composite exceeded the GRPO anchor by $+22.66/+22.37$ and $+11.72/+11.59$ points, but all three failed the frozen $+2/+2$ gate against SFT; none was scaled. Exact episode coverage, zero inference errors, five paired exports, and independent metric checks were verified; a final raw-ID audit checked 1,024 evaluator records. These are single-configured-seed results on reused training scenes, not a clean generalization test. Turn-level feedback uses privileged simulator geometry; the SRGPO-style result is a composite that also changes outcome normalization, so it does not isolate process grouping.

Sequence-level policy-loss objectives. We next tested three sequence aggregation objectives from the same navigation SFT initialization: GSPO-style sequence likelihood, GMPO-style geometric-mean likelihood, and arithmetic token/sequence averaging. Each used 64 group-four updates on the same 512 fit rows with configured seed 11. On the adaptively reused 256-episode, eight-scene development screen, the candidates achieved 53, 34, and 46 successes, respectively, versus 109 for matched FP16 SFT. Paired SR/SPL changes were $-21.88/-20.56$, $-29.30/-27.98$, and $-24.61/-23.29$ points. All three scene-bootstrap intervals were below zero for both metrics, and every method failed the frozen $+2/+2$-point gate. The raw-ID audit checked 1,024 evaluator records; exact coverage, zero inference errors, and an independent compact recount were confirmed. No method was scaled. These are single-seed results on adaptively reused training scenes, not a clean generalization test.

Online shortest-path action correction. We tested an on-policy DAgger-style auxiliary loss using one privileged shortest-path primitive action at a sampled state from each trajectory. The group-four candidate added token-masked action cross-entropy to turn-level RLOO; its matched control used the same reward and optimizer without that loss. Both ran 64 updates from the same navigation SFT initialization. On the fixed, adaptively reused 256-episode, eight-scene development screen, matched FP16 SFT achieved 109 successes (SR/SPL 42.58/41.26%), the RLOO control 52 (20.31/20.31%), and the DAgger candidate 0 (0/0%). Every arm covered 256 unique episodes with zero inference errors, and independent raw-stat and compact recounts agree. Candidate-minus-SFT paired SR/SPL were $-42.58/-41.26$ percentage points, far below the frozen joint $+2/+2$ gate; the method was stopped without scale-up. The teacher is privileged simulator supervision, not a deployable semantic signal. This is one configured-seed result on an adaptively reused development split, not a clean generalization test. Teacher-query latency was not separately measured. The original post-check markers and output-path recovery are archived; no inference was repeated.

Table 7. Exploratory 128-step policy-prefix follow-up on all 1,839 val-unseen episodes. Differences are branching minus a same-data, from-scratch destination-only control, in percentage points. Mean ± sample standard deviation is over three training seeds; each model has one stochastic decode per episode.

Seed	Δ SR	Δ SPL
11	−7.18	−6.86
22	+0.49	+0.80
33	+3.26	+4.41
Mean ± SD	-1.14 ± 5.41	-0.55 ± 5.75

D. Start-state and rollout-count comparisons

Exploratory policy-prefix follow-up. Separately from EventTrace, we tested whether changing the training-state distribution helps destination-only GRPO. For each of three seeds, a branching curriculum replayed a training-only policy prefix before sampling two continuations; its matched control started at the task origin. Both arms used the same 512 train episodes, SFT initialization, destination reward, and 128 optimizer steps. All six checkpoints covered the complete 1,839-episode val-unseen split with zero inference errors. Table 7 shows sharply different seed effects and a negative three-seed mean. Exploratory scene-and-seed 95% intervals for the mean differences are $[-6.51, 4.02]$ SR and $[-6.16, 4.85]$ SPL points. On the 1,583 episodes outside the fixed 256-episode development screen, the mean changes remain −1.22 SR and −0.55 SPL points. Separate stochastic decodes of the overlapping 256 episodes changed success outcomes for 20–41 episodes per checkpoint, so the small screen is an unstable model-selection signal. This follow-up changes start state and remaining budget together; it does not identify why EventTrace failed or establish a benefit from branching. The checked paired episode records and training audits accompany the manuscript.

Group-four optimizer follow-up. A separate destination-only comparison tests four sampled trajectories per GRPO group against the same-data, two-sample control. Both arms use the same SFT initialization, 512 unique training episodes, and 128 optimizer steps at each of three matched seeds; the four-sample arm uses twice as many simulator rollouts. All six checkpoints cover the identical 1,839-episode val-unseen manifest with zero inference errors. Table 8 shows positive paired differences in each seed: the mean is $+2.10 \pm 1.93$ SR and $+2.51 \pm 2.27$ SPL percentage points (mean ± sample standard deviation over seeds). On the 1,583 episodes outside the fixed 256-episode screen, the means are +1.94 SR and +2.40 SPL points. The largest seed-33 gain is +4.30 SR points, while exploratory 11-scene bootstrap intervals for seeds 11 and 22 cross zero. One stochastic decode per checkpoint and only three training seeds limit the inference. This comparison does not identify a better semantic reward; subsequent reward comparisons must hold group size at four in both arms. Paired records, training audits, and independent recount code accompany the manuscript.

Table 8. Exploratory 128-step, destination-only group-size follow-up on all 1,839 val-unseen episodes. Differences are four samples minus a matched two-sample control, in percentage points; the four-sample arm uses twice the simulator rollouts.

Seed	Δ SR	Δ SPL
11	+0.65	+0.96
22	+1.36	+1.44
33	+4.30	+5.12
Mean ± SD	$+2.10 \pm 1.93$	$+2.51 \pm 2.27$

E. Representation-to-reward development

Instruction-grounded reward representation. We also probe a potential reward that combines a causal progress encoder from frozen navigation-policy states with a SigLIP image–instruction encoder adapted on training-scene goal panoramas and group-four successful/failed routes. SigLIP text is padded to a fixed 64 tokens, and offline scoring uses the same one-image request shape as the online service. Each component is normalized by its fit-scene mean absolute margin; a fixed 1:1 mixture is applied without development-set tuning. On 52 development pairs, successful-route ranking rises from 37/52 for the temporal term and 32/52 for the visual term to 43/52 for the mixture; correct-versus-wrong-instruction preference is 39/52. On 38 new episode-disjoint, train-scene pairs, the mixture scores 30/38 route ranks and 34/38 instruction comparisons. A previously inspected 48-pair scene audit scores 41/48 and 37/48. These selected offline pairs share train scenes or prior audit use, so they provide a screen for an online group-four reward comparison, not evidence of val-unseen navigation gain.

Online fused-reward screen. We added this frozen potential as a bounded terminal bonus to the destination reward and trained a matched group-four policy for 64 steps at seed 11. Its training audit verifies 1,024 reward requests for 1,024 rollouts across 256 matched train episodes. On the fixed 256-episode val-unseen screen, the fused-reward checkpoint achieves 68/256 successes versus 80/256 for the same-data, same-group-

size destination-only control: paired SR changes by −4.69 percentage points and SPL by −4.34 points. Both arms cover exactly 256 episodes in 11 scenes with zero inference errors. This single-seed screen does not support scaling this particular reward, despite its offline ranking results. A failure-aware encoder trained on unsuccessful pairs improves a separate 106-pair train-scene ranking audit from 69/106 to 79/106, but direct fusion with the image–instruction score weakens reused instruction-matching probes. These findings do not establish a navigation gain.

Failure-only reward screen. We next applied the new temporal potential only to unsuccessful rollouts, leaving successful rollouts on the original destination reward. A matched seed-11 group-four policy completed 64 steps on the same training episodes; its audit found 854 reward requests for 854 unsuccessful rollouts and no extra bonus for 170 successful rollouts. On the same fixed 256 val-unseen episodes, it achieves 63/256 successes versus the destination-only control's 80/256: paired SR changes by −6.64 points and SPL by −6.05 points. Both models cover 11 scenes without inference errors. The exploratory scene-cluster 95% intervals are $[-11.07, -2.29]$ SR and $[-10.66, -1.47]$ SPL points. This screen fails the predeclared positive SR-and-SPL scale gate; we therefore do not run a three-seed scale-up for this reward. Training-only failed-pair ranking (622/970 in all-failure groups) does not transfer into a navigation benefit. A post-hoc behavior check finds maximum-turn termination on 187 candidate episodes versus 107 control episodes, while mean traveled path length falls from 6.74 to 5.71 m; these differences motivate a stopping-aware reward test but do not identify the cause of the regression. An offline equal mixture with the adapted visual score also fails its train-scene development gate (68/125 failed-pair ranks versus 90/125 for the temporal term), so its audit remains unopened.

Stop-aware reward screen. To test whether terminal-mode conditioning mitigates this failure, we gave the frozen temporal bonus only to unsuccessful voluntary STOP trajectories. Successes retained the destination reward, while turn-limit and format failures received no bonus. The matched seed-11 group-four policy completed 64 steps on the same training episodes. Its audit found exactly 328 scorer requests for 328 unsuccessful voluntary stops across 1,024 rollouts, with nonzero actor gradients at every step. On a new, disjoint 256-episode val-unseen screen covering 10 scenes, the candidate succeeds on 45/256 episodes versus 93/256 for its same-seed destination-only control. Paired SR and SPL changes are −18.75 and −17.43 percentage points; both arms have zero inference errors. The predeclared positive SR-and-SPL gate failed, so this reward received no three-seed expansion. Post-hoc analysis finds mean path length 4.44 versus 6.93 m and only three candidate episodes with a non-null termination reason versus 116 controls, consistent with frequent early stops. On the candidate's train rollouts, its frozen raw score has AUC 0.486 for distinguishing 18 unsuccessful STOP endpoints within 3.5 m of the goal from 310 farther endpoints; only 11 of 183 positive-bonus stops are within 3.5 m. This train-scene diagnostic does not establish causality, but it shows that failure-pair ranking is insufficient calibration for rewarding STOP.

Mode-stratified ordinal screen. To remove a direct reward offset between unsuccessful STOP and turn-limit trajectories, we next ranked frozen temporal scores only within the same terminal mode of each four-rollout all-failure group; any-success groups retained the original outcome reward. This is a trajectory-level group comparison, not true turn-wise credit assignment: the current GRPO implementation broadcasts a group-normalized total reward to all action tokens. The 64-step seed-11 training audit verified 1,024 rollouts, 149 all-failure groups with active ordinal rewards, 827 scorer requests for exactly 827 failures, and a nonzero actor update at every step. On a third fixed, nonoverlapping 256-episode val-unseen screen spanning nine scenes, the mode-stratified candidate succeeds on 59/256 episodes versus 73/256 for its same-data, same-seed group-four outcome-only control. Paired SR and SPL change by −5.47 and −5.23 percentage points, with zero inference errors in either arm. The exploratory nine-scene bootstrap 95% intervals are $[-9.06, -1.95]$ SR and $[-8.98, -1.74]$ SPL points. Among failures, the candidate ends with an unforced model STOP 148 times versus 84 for the control; this post-hoc association does not identify a causal mechanism. The predeclared positive SR-and-SPL scale gate failed, so no 128-step or three-seed extension was run for this reward. Absolute success rates across the three 256-episode screens are not directly comparable because their episode sets differ.

Privileged turn-wise credit upper bound. The preceding trajectory-level rewards broadcast one return to every action token. To isolate whether finer credit assignment can help, we tested a training-only oracle that labels each executed turn by its change in simulator geodesic distance. In all-failure groups of four, it centers return-to-go among active rollouts at each turn and gives no auxiliary advantage to generated STOP

Table 9. Matched group-four, 512-row, 128-step privileged turn-wise oracle versus destination-only control on all 1,839 val-unseen episodes. Differences are oracle minus control in percentage points.

Seed	SR counts O/C	Δ SR	Δ SPL
11	513/450	+3.43	+3.45
22	525/545	−1.09	−0.61
33	585/492	+5.06	+4.70
Mean ± SD	—	$+2.47 \pm 3.18$	$+2.51 \pm 2.78$

tokens; groups containing a success retain destination-outcome GRPO. Geodesic distance is never a policy observation, but this oracle is not a learned semantic reward. The same-data seed-11 pilot completed 64 steps. An independent training audit checked 256 four-rollout groups, 11,450 turns, active contrast in all 141 all-failure groups, and nonzero actor gradients on all 64 steps. On a fourth, pre-frozen 256-episode val-unseen screen spanning nine scenes, it succeeds on 81 episodes versus 72 for its outcome-only control: paired SR +3.52 and SPL +3.76 percentage points, with zero inference errors. The exploratory scene bootstrap 95% intervals span $[-2.73, 9.13]$ SR and $[-2.58, 9.28]$ SPL points. This one-seed screen selected a same-checkpoint full val-unseen recheck. Across all 1,839 episodes, the oracle succeeds on 602 versus 545 for the control: paired SR +3.10 and SPL +3.05 points. On the 1,583 episodes outside the reused screen, it succeeds on 521 versus 481: SR +2.53 and SPL +2.50 points. Both evaluations have exact coverage and zero inference errors. This remains a post-screen, one-seed development result using privileged training labels; it neither validates EventTrace's semantic verifier nor establishes a deployable learned reward.

We next scaled this privileged mechanism to 512 distinct training episodes and 128 optimizer steps at each of three seeds, with four rollouts per group in both oracle and destination-only arms. All six checkpoints were decoded once on the same 1,839 val-unseen episodes, with exact coverage and zero inference errors. Table 9 gives paired results. The mean oracle change is $+2.47 \pm 3.18$ SR and $+2.51 \pm 2.78$ SPL points (sample SD over seeds); seed 22 instead loses 1.09 SR points. On the 1,583 episodes outside the reused 256-item development screen, the means are +2.63 SR and +2.64 SPL points. Exploratory scene-and-seed intervals for the full-set mean are $[-1.49, 6.05]$ SR and $[-1.20, 5.78]$ SPL points. Thus the observed mean is positive, but the seed variation and intervals do not support a consistent gain. The reward uses simulator distance.

An adaptive stopping-boundary pilot tested whether capping geodesic progress credit inside the 3 m success region and penalizing further movement there by 0.1 improves this oracle. Its seed-11 matched n=4 64-step audit verified 256 identical training rows, nonzero actor gradients at every step, and 902 inside-boundary penalty turns. On the reused 256-episode development screen, the candidate succeeds 67 times versus 72 for the unchanged control: paired SR −1.95 and SPL −1.62 points, with zero inference errors. This fails the fixed positive SR-and-SPL scale gate. Neither privileged variant establishes a deployable semantic reward.

Two observation-only progress representations were screened before online use. An instruction-conditioned scalar potential fit to cached policy RGB histories reached only 48.88% balanced forward/regression accuracy on its small train-scene development split. A joint two-view antisymmetric LoRA reached 53.52% balanced accuracy and 52.08% correct-versus-wrong-instruction preference at its selected checkpoint; at 9.88% stationary false positives, its forward recall was zero. Neither passed the fixed development gate or entered RL. A further start-anchored, action-conditioned three-view LoRA also failed its small development gate (55.51% balanced direction, 22.22% regression accuracy), so it received no audit or online reward test. This leaves distillation of the privileged credit signal into an observation-only representation as open work.

We also screened arrival evidence on a scene-disjoint R2R-train split of audited group-four policy rollouts. A single-frame occupancy head reached AUC 0.6053, but recalled only 8.33% of inside-region views and 5.04% of inside-region views on unsuccessful routes at a pooled false-positive rate below 5%. A separate joint two-view, instruction-conditioned transition head was trained for a fixed 1,024 microsteps on real crossings, far nonarrivals, real near-goal retreats, and identical-view wrong-instruction pairs. On all 758 development pairs from 351 records, its selected checkpoint reached AUC 0.6430, versus 0.5177 for a text-only control and 0.5211 after same-scene image-pair shuffling. For identical image pairs with different instructions, the correct instruction ranked higher in only 98/176 cases (55.68%). A CPU-only source audit confirms that these pairs share the exact simulator states; their wrong-instruction labels are geometric alternate-goal proxies, without independent semantic adjudication. Weak grounding and label ambiguity cannot be separated by this diagnostic. Nevertheless, at 4.89% pooled false positives it recalled only 15.63% of crossings and 13.45% of unsuccessful crossings, below the frozen 55% and 50% gates. These exploratory screens reused development scenes across method design; neither opened its reserved train-scene audit or entered online RL. Better AUC therefore supplies no evidence

of a navigation gain from this learned reward.